Leveraging Physical Rules for Weakly Supervised Cloud Detection in Remote Sensing Images

Yang Liu, Qingyong Li, *Member, IEEE*, Xiaobao Li, Shuyi He, Fengjiao Liang, Zhigang Yao, Jun Jiang, and Wen Wang

***Abstract*— Cloud detection plays a significant role in remote sensing (RS) image applications. Existing deep learning-based cloud detection methods rely on massive precise pixelwise annotations, which are time-consuming and expensive. To alleviate this problem, we propose a weakly supervised cloud detection framework that leverages physical rules to generate weak supervision for cloud detection in RS images. Specifically, a rule-based adaptive pseudo labeling (RAPL) algorithm is devised to adaptively annotate potential cloud pixels based on cloud spectral properties without manual intervention. Unlike existing physical annotations using fixed thresholds, RAPL employs the bidirectional threshold segmentation and adaptive gating mechanism to annotate cloud and boundary masks with more explicit semantic categories and spatial structures separately. Subsequently, these pseudo masks are treated as weak supervision to optimize the heuristic cloud detection network for pixelwise segmentation. Considering that clouds appear as complex geometric structures and nonuniform spectral reflectance, a deformable boundary refining module is designed to enhance the modeling ability of spatial transformation and activate sharp boundaries from translucent cloud regions. Moreover, a harmonic loss is employed to recognize clouds with nonuniform spectral reflectance and suppress the interference of bright backgrounds. Extensive experiments on the GF-1, L8 Biome, and weakly supervised cloud detection (WDCD) datasets demonstrate that the proposed method achieves state-of-the-art results. A public reference implementation of this work in PyTorch is available at https://github.com/NiAn-creator/HeuristicCloudDetection.**

***Index Terms*— Cloud detection, convolutional neural network (CNN), remote sensing (RS) image, weakly supervised semantic segmentation.**

I. Introduction

With the advancement of aerospace technology, remote sensing (RS) images are increasingly being used in various research works, such as land cover monitoring [1], [2], [3] and environmental protection [4]. However, over 67% of the Earth's surface is covered by clouds [5], making most RS images would inevitably contain cloud pixels. On the one hand, the cloud is a vital meteorological element that contributes to atmospheric observation by providing valuable information for weather forecasting [6] and climate prediction [7]. On the other hand, cloud cover impacts various RS image applications by obstructing satellite sensors from capturing clear views of the Earth's surfaces. Therefore, it is crucial to accurately distinguish clouds from clear pixels in RS images to maximize data availability. It facilitates the utilization of RS images in various applications, such as photovoltaic power generation [8] and airline traffic safety [9]. Moreover, spaceborne cloud detection can proactively eliminate low-quality images severely disturbed by clouds [10], [11], thereby improving the efficiency of satellite data transmission. Consequently, cloud detection has played an essential role in RS image applications and has been a popular research topic in recent decades [12], [13], [14].

The objective of cloud detection is to separate cloud and clear pixels in RS images. Earlier works mainly rely on the physical properties of clouds and employ spectral differences to differentiate clouds from the background [15], [16], [17]. These algorithms typically achieve high overall accuracy (OA) by applying combined spectral filters. However, as satellite imaging resolution increases, RS images contain more detailed spatial information. It has become challenging to perform cloud detection based solely on spectral characteristics in complex scenes, such as cloud-snow coexistence or thin cloud detection [18]. Hence, spatial features, such as geometric structures and textures, are considered [19], [20].

Inspired by the remarkable progress of deep learning in computer vision, the convolutional neural network (CNN) has been widely investigated for cloud detection and achieved excellent performance [21], [22], [23], [24], [25], [26]. These methods conduct automatic feature extraction and cloud detection via establishing a complex nonlinear mapping between input and output. However, most of these approaches require massive precise pixelwise annotations, which are time-consuming and expensive since the irregular geometric structure and nonuniform spectral reflectance of clouds.

To alleviate the annotation burden, weakly supervised cloud detection (WDCD) has received extensive attention. One mainstream approach applies patch labels that only indicate the presence or absence of clouds in RS image patches [27], [28],

Manuscript received 8 December 2022; revised 6 May 2023 and 16 June 2023; accepted 5 July 2023. Date of publication 12 July 2023; date of current version 27 July 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62276019 and Grant 62006017, and in part by the Fundamental Research Funds for the Central Universities through the Science and Technology Leading Talent Team Project under Grant 2022JBQY007 and Grant 2023JBZY037. *(Corresponding author: Wen Wang.)*

Yang Liu, Xiaobao Li, Shuyi He, Fengjiao Liang, and Wen Wang are with the Key Laboratory of Big Data and Artificial Intelligence in Transportation, Beijing Jiaotong University, Beijing 100044, China (e-mail: yliucit@bjtu.edu.cn; wangwen@bjtu.edu.cn).

Qingyong Li is with the Key Laboratory of Big Data and Artificial Intelligence in Transportation, Ministry of Education, and the Frontiers Science Center for Smart High-speed Railway System, Beijing Jiaotong University, Beijing 100044, China.

Zhigang Yao and Jun Jiang are with the Beijing Institute of Applied Meteorology, Beijing 100029, China.

Digital Object Identifier 10.1109/TGRS.2023.3294817

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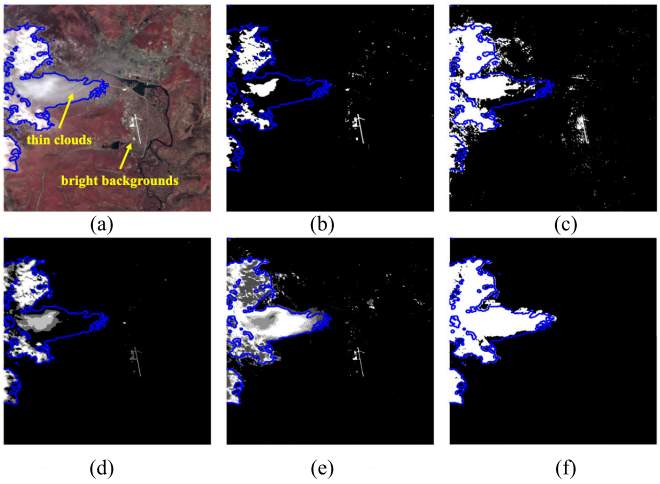


Fig. 1. Examples of dilemmas in cloud detection. (a) Translucent thin clouds and bright ground objects pose challenges to cloud detection (L8 Biome image). The blue line depicts the cloud contour in ground truths. (b) Fmask is a classic physical algorithm with globally optimal thresholds. (c) WDCD uses patch labels for cloud detection. (d) and (e) Pseudo masks annotated by RAPL (cloud mask and BM, respectively). (f) HCDNet activates entire cloud contours and suppresses bright backgrounds.

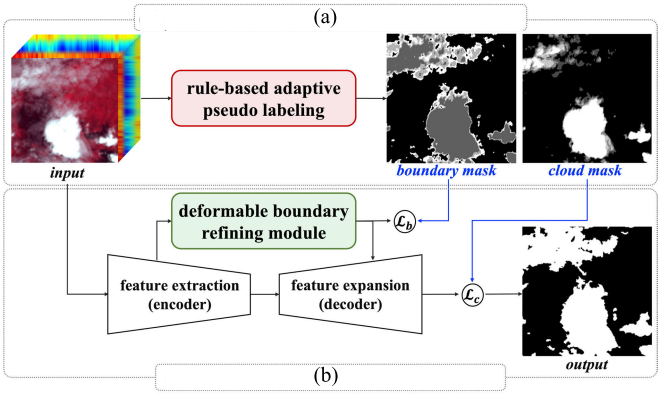


Fig. 2. Pipeline of the proposed WDCD framework. Given an input image, RAPL is devised first to annotate pseudo masks based on cloud spectral properties without manual intervention. Next, these annotated pseudo masks are treated as weak supervision to optimize the HCDNet for pixelwise segmentation. (a) Weak annotation using the physical rule. (b) HCDNet.

[29], [30], [31]. Typically, an image classifier is first trained to generate cloud activation maps, which are then segmented into pixelwise cloud masks through statistical thresholds [27], [28]. Another strategy uses generative models, such as generative adversarial networks, to separate clouds and backgrounds [29], [30], [31]. These models aim to learn image-to-image translations between clear and cloudy images, where only cloud pixels are affected during translation. This allows for deriving pixelwise cloud masks based on the difference between the original and translated images. However, since patch annotations only provide global categories, these methods tend to capture global semantic information while ignoring detailed spatial structures. As shown in Fig. 1, WDCD [27] is a typical cloud detection model that uses patch labels. WDCD performs well in detecting opaque clouds but faces challenges in detecting translucent thin clouds and excluding bright ground objects.

Along with patch labels, many researchers have also utilized traditional physical algorithms to generate pseudo masks for WDCD. For instance, DeepCloud [32] erodes the prediction of Fmask [16] as weak supervision during model training. However, the thresholds of physical algorithms are selected empirically and are globally optimal, which might introduce local errors. As shown in Fig. 1, there are evident omission errors of thin clouds and false alarms of bright backgrounds in the Fmask prediction.

To tackle these problems, we propose a novel WDCD framework that leverages physical rules to generate weak supervision adaptively for cloud detection in RS images. As shown in Fig. 2, this framework mainly contains two components, i.e., a weak annotation using the physical rule and the heuristic cloud detection network (HCDNet). Specifically, a rule-based adaptive pseudo labeling (RAPL) algorithm is devised to generate pseudo annotations adaptively according to cloud spectral properties without manual intervention. Unlike general physical annotations [32], RAPL employs bidirectional threshold adjustment and adaptive gating mechanism to annotate cloud mask and boundary mask (BM) with more explicit semantic categories and spatial structures separately. These generated pseudo masks are then taken as weak supervision to optimize HCDNet for pixelwise cloud detection. Considering that prevailing standard CNN is inadequate to represent the complex geometric structure of clouds through regular sampling grids [21], [22], a deformable boundary refining module (DBRM) is designed to improve the modeling ability of spatial transformation and activate precise boundaries from translucent cloud regions. Furthermore, a harmonic loss is developed to enhance the perception of clouds with nonuniform spectral reflectance and suppress the interference of bright backgrounds.

The main contributions of this work are summarized as follows.

1) We explore a WDCD framework that leverages the physical rule to annotate pseudo masks with explicit semantic and spatial guidance for model training without manual intervention.
2) We present a DBRM to learn the complex spatial transformation of clouds and activate translucent clouds with sharp boundaries. Meanwhile, a harmonic loss is designed to recognize clouds with nonuniform spectral reflectance and exclude bright backgrounds.
3) Extensive experiments on the GF-1 [19], L8 Biome [33], and WDCD [27] datasets demonstrate the effectiveness of our proposed method and establish state-of-the-art results.

The remainder of this work is arranged as follows. Section II briefly reviews related works. Section III specifies the proposed method. Section IV shows datasets and experiments. Discussion is provided in Section V. Finally, Section VI summarizes the proposed framework.

II. Related Work

This section briefly reviews relevant works that include cloud detection in RS images and weakly supervised semantic segmentation for RS images.

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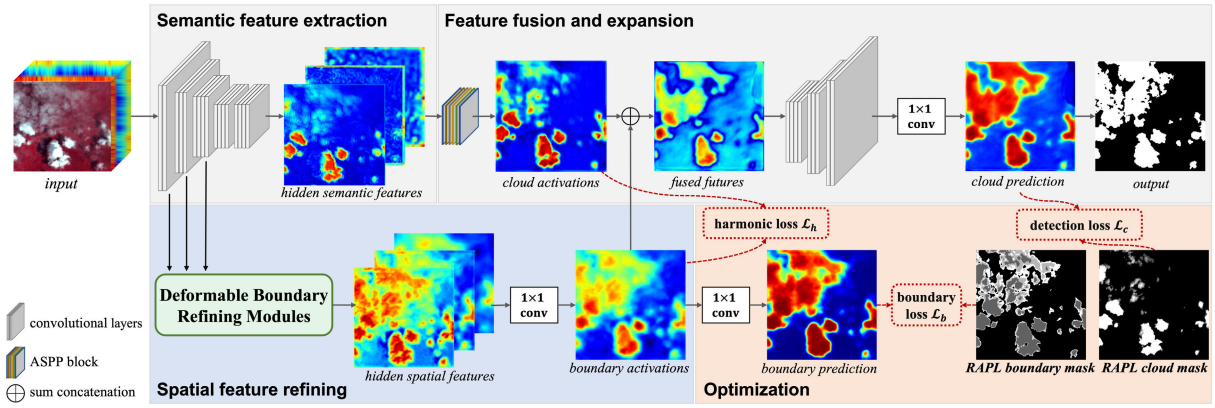


Fig. 3. Overall architecture of the HCDNet. HCDNet first uses an encoder to extract semantic features. Then, DBRMs are applied in shallow layers to refine spatial features. Next, these hidden features are processed to generate cloud and boundary activations, which are then concatenated to facilitate precise cloud prediction. The cloud mask and BM annotated by RAPL are used to optimize HCDNet training. A harmonic loss is designed to diversify cloud and boundary activations while constraining identical category identification.

A. Cloud Detection in RS Images

Cloud detection plays an essential role in RS image applications and has emerged as a popular research topic in recent decades. Based on the involved technique, existing cloud detection research can be roughly categorized as physics-based methods and learning-based methods.

Physics-based methods utilize the physical properties of clouds to create discriminative features and identify appropriate thresholds for cloud detection. From the feature perspective, earlier algorithms mainly combine multiple spectral rules to segment clouds by thresholding [15], [16], [17], [34], [35]. For example, Zhu and Woodcock et al. [16] and Zhu et al. [17] combined temperature probability, spectral variability, and brightness probability to produce cloud masks. With the increasing availability of high-resolution satellites, RS images comprise more detailed spatial information. It is challenging to detect clouds based solely on spectral features. As a result, researchers have incorporated spatial features such as geometry and textures to improve cloud detection results [19]. In addition, some studies have fused multitemporal spectral or spatial variations to boost the accuracy [36], [37]. While these traditional methods have proven effective, there are still some limitations. On the one hand, the selection of physical rules is empirical and strict on the input spectral bands, leading to limited generalization ability. On the other hand, the globally optimal threshold setting cannot consider cloud objects under different scenes, which inevitably leads to local errors.

Motivated by the progress of machine learning, traditional machine learning algorithms based on handcrafted features have been applied to cloud detection and achieved continued improvements, such as the support vector machine [38] and k-means [39]. Recently, deep CNNs have made significant achievements in cloud detection due to their powerful learning capabilities [21], [22], [23], [24], [25], [26]. Yang et al. [26] applied category correlation and knowledge distillation to cope with thin clouds and cloud-like objects in complex scenes. Ji et al. [25] combined multiscale features and proposed channel attention to seek global consistency. Luo et al. [10] proposed a lightweight network to alleviate computational resources and applied it to airborne cloud detection. However, these methods depend on numerous pixelwise annotations, which are laborious and expensive.

The WDCD has gained increased attention due to its more convenient labeling. The commonly used weak supervision includes patch labels [27], [28], [29], [30], [31] and scribbles [32]. Specifically, patch labels indicate the presence or absence of clouds in RS image patches. Under patch supervision, many researchers frame cloud detection as a semantic segmentation task [27], [28]. Li et al. [27] proposed a binary classifier to identify cloudy and clear patches and generate cloud activation maps, which are then segmented into pixel-level cloud masks by statistical thresholding. Others perform cloud detection via generative models [29], [30], [31]. Zou et al. [29] treated cloud detection as a mixed energy separation task. They first extracted the cloud components of a cloudy image by generative networks and then fused them with a clear image to obtain a fake cloud image. Next, these cloud components were used as reference masks to train a matting network for cloud detection. As to scribble supervision, Molinier et al. [32] eroded the prediction of Fmask as weak supervision to optimize model training. In addition, some works introduced multitemporal [40] or multisource data [41] to extract auxiliary information for cloud detection.

Unlike objects in natural images, clouds appear as complex geometric structures with irregular boundaries, varying from fiber to lump. This brings significant challenges for accurate cloud detection, especially around boundaries. Researchers have made great efforts to enhance boundary localization accuracy in recent years. Liu et al. [23] used deformable convolutional layers to enhance the modeling ability of spatial transformation and facilitate sharp boundaries. Peng et al. [42] found that reducing the network receptive field could enhance the boundary accuracy of cloud detection. Alternatively, He et al. [22] designed a boundary loss to direct the network to focus on cloud boundary information and optimize the relevant detection results. Guo et al. [43] employed a spatial attention module to fuse multilevel features seamlessly and optimize the boundary blur. Different from the above methods,

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postprocessing strategies have been adopted to refine boundary detection results, such as the conditional random field [24], [44]. Inspired by these, we produce DBRMs in shallow layers of HCDNet to capture detailed spatial features and activate precise boundaries among translucent cloud regions.

B. Weakly Supervised Semantic Segmentation for RS Images

Semantic segmentation is a fundamental computer vision task that aims to predict a semantic category for each pixel in an image. Since AlexNet [45] won the ILSVR contest in 2012, the CNN-based semantic segmentation model has been widely adopted in the RS community and performed with impressive accuracy [46]. Yet, compared with other vision tasks such as classification and object detection, semantic segmentation needs to collect dense pixelwise labels, which are time-consuming and expensive.

In order to reduce the annotation burden, researchers have been exploring weakly supervised semantic segmentation techniques for RS images that use more convenient pseudo masks. Generally, the pseudo masks include patch labels [47], [48], scribbles [49], bounding boxes [50], and points [51], [52]. Zhou et al. [47] used patch tags to determine the approximate location of landslides and combined them with a generative adversarial network to segment the landslide area. Wei and Ji [49] proposed a buffer-based strategy to propagate scribbles for road surface extraction. Rafique and Jacobs [50] presented a generative probabilistic mask to predict dense building masks via bounding box annotations. Lu et al. [51] studied a neighbor feature aggregation network that can utilize point annotations to extract water bodies.

Unlike semantic segmentation in natural images, RS images comprise more detailed spatial and spectral information. The weakly supervised semantic segmentation for RS images should pay more attention to these specific properties of objects. As a result, this work leverages the physical rule based on cloud spectral characteristics to annotate potential cloud mask and BM with more specific semantic categories and spatial contours separately for model training.

III. Methodology

This work aims to leverage a simple physical rule to annotate pseudo masks at minimum cost and then treat them as weak supervision to learn segmentation features for cloud detection during training. Sections III-A–III-D start with an overview of the proposed framework and then detail the designed modules.

A. Overview

As shown in Fig. 2, the proposed WDCD framework comprises two components: 1) weak annotation using the physical rule and 2) HCDNet. In detail, an RAPL algorithm based on cloud spectral properties is devised to generate pseudo cloud mask and BM without manual intervention. Considering that the fixed initial threshold is inadequate to annotate clouds with nonuniform spectral reflectance in complex scenes, RAPL performs a bidirectional threshold segmentation to mark precise opaque cloud pixels and potential translucent boundaries or thin clouds. Then, an adaptive gating mechanism is developed to select valid annotations for pseudo labeling. Subsequently, these adaptive annotated cloud mask and BM are treated as ground truths to optimize HCDNet training.

Fig. 3 shows the overall architecture of HCDNet. An input image is first fed into a replaceable encoder (e.g., the ResNet-34 [53]) to extract semantic information and yield multilevel features. Unlike the object in natural images, clouds appear as complex geometric structures and nonuniform spectral reflectance. Under weak supervision, HCDNet is prone to capture more discriminative semantic features but ignores spatial details, giving sparse cloud activations. As shown in Fig. 3, cloud activation maps only emphasize opaque clouds while neglecting translucent boundaries and thin clouds. To tackle this problem, DBRMs are developed in shallow layers to enhance the transformation modeling capability of HCDNet and refine spatial features through irregular sampling structures. Subsequently, a 1×1 convolutional layer is used to derive boundary activations.

At the feature fusion and expansion branch, an atrous spatial pyramid pooling module [54] is applied to hidden semantic features to incorporate multiscale contexts and generate cloud activation maps. Next, cloud and boundary activation maps are concatenated and fed to three consecutive decoder blocks to gradually restore spatial resolution. Each decoder block includes convolutional layers symmetrical to the encoder and a bilinear upsampling layer. The dense cloud prediction can be obtained by a 1×1 convolutional layer and normalized to 0–1 by the sigmoid.

Finally, we use a threshold ξ to segment binary cloud detection results. A pixel can be predicted as a cloud if its probability value exceeds ξ.

B. Rule-Based Adaptive Pseudo Labeling

While superficially resembling natural images, clouds have complex geometric structures and nonuniform spectral reflectance that gradually fades from the center to the edge. This makes it challenging to perform pixelwise cloud annotation by hand in RS images. Even though physical cloud detection algorithms can automatically generate cloud masks, those segmentation thresholds are global optima, leading to potential local errors [18], [19]. As shown in Fig. 1, Fmask ignores transparent boundaries and misclassifies the bright ground objects. To address this issue, an RAPL algorithm is devised to leverage the physical rule for pseudo annotation adaptively.

Previous research has found that clouds always hold higher reflectivity than clear sky pixels in visible bands, especially in blue and red bands where their spectral response differs from that of most land surfaces [34], [35]. Inspired by this, RAPL employs a spectral reflectance rule based on haze optimized transformation (HOT) [34] to capture pure cloud regions. Concretely, HOT is a classic physical rule and has proven its effectiveness in various satellite images [16], [17]. The HOT spectral reflectance rule used in RAPL is formulated as

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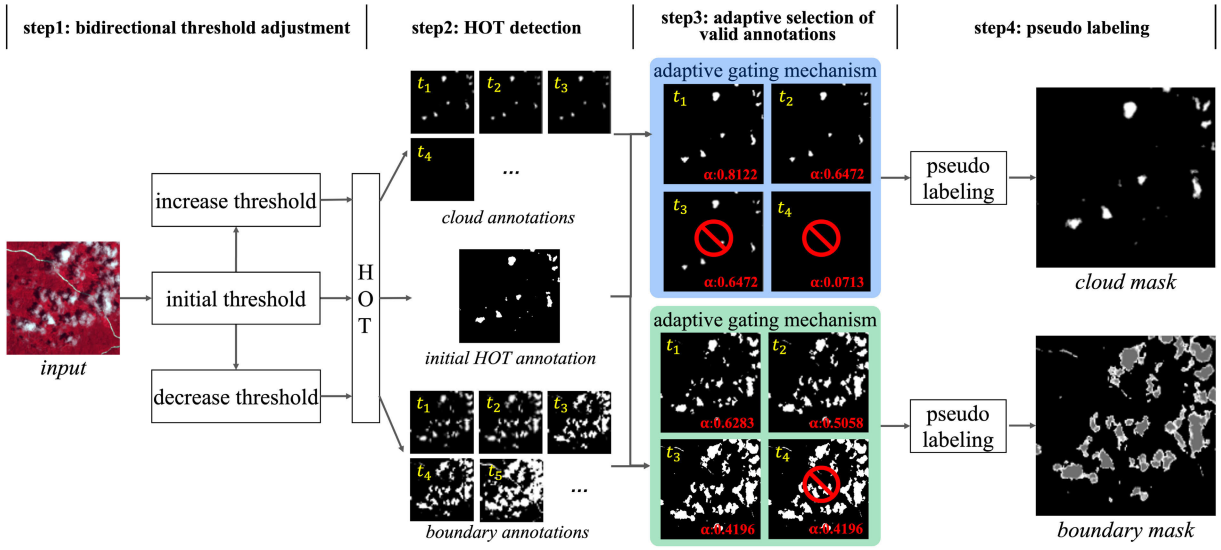


Fig. 4. Illustration of the RAPL. RAPL first performs bidirectional threshold adjustments to annotate potential cloud pixels. Then, an adaptive gating mechanism is applied to select valid annotations. The cloud mask and BM can be derived accordingly. The yellow t_i in annotations means the ith adjusted threshold. The red α represents the affinity between the current annotation and the initial HOT annotation. The red circle with a 45° diagonal line means the prohibition sign, indicating that the current annotation is invalid and should be discarded.

follows:

$$\text{HOT} = \text{Blue} - 0.5 \times \text{Red} \tag{1}$$

where Blue and Red denote the blue and red band reflectance, respectively. Before executing this rule, the digital number value of spectral bands is converted to the top of atmosphere reflectance to reduce between-scene variability [19], [55]. Then, an empirical initial threshold t_{init} based on classic algorithm settings is constructed to separate cloud and clear pixels. For instance, $t_{\text{init}} = 0.13$ in MFC [19]. However, the initial threshold cannot adequately consider the full extent of clouds with nonuniform spectral reflectance. As shown in Fig. 9, the HOT initial threshold emphasizes opaque clouds while remaining insensitive to translucent clouds, and there are some false alarms with bright buildings in the coast and urban area scenes.

To alleviate these local errors, RAPL produces the bidirectional threshold adjustment to annotate pseudo cloud mask and BM with more specific semantic categories and spatial structures separately. As shown in Fig. 4, RAPL first adjusts the initial threshold t_{init} with step size s from two directions for more accurate cloud identification or more complete cloud contours. The adjusted threshold can be defined as follows:

$$t_k = t_{k-1} + d \times k \times s \tag{2}$$

where $k = [1, 2, \ldots, K]$ indicates the times of adjustments. In particular, $t_0 = t_{\text{init}}$. $d \in \{-1, 1\}$ is a direction indicator that defines decreasing or increasing the threshold. s denotes the step size. Thus, given a threshold t_k, if the HOT value of a pixel is bigger than t_k, this pixel will be marked as a cloud; otherwise, it will be marked as a clear sky. The HOT annotation $\mathcal{A}_{t_k} \in \mathbb{R}^{H \times W}$ can be defined as follows:

$$\mathcal{A}_{t_k} = \text{HOT} > t_s, \quad s \in K \tag{3}$$

where the operation ">" denotes a pixelwise spectral reflectance comparison and H and W are the height and width of the input, respectively.

It is worth noting that the HOT rule is a binary linear threshold segmentation. When the threshold increases, the activation area shrinks accordingly and vice versa. Therefore, increasing the threshold in HOT rule allows for the annotation of pixels with higher spectral reflectance, which suggests more specific semantic categories. On the contrary, decreasing the threshold enables the annotation of translucent boundaries and thin clouds, which indicates more complete spatial structures. For convenient description, we denote the annotations labeled by increasing thresholds as cloud annotation $\mathcal{A}^c_{t_k}$, and the annotations labeled by decreasing thresholds as boundary annotation $\mathcal{A}^b_{t_k}$.

Due to only incorporating the red and blue bands, HOT detection cannot adequately suppress bright surfaces and so often overestimate cloud thicknesses over bright backgrounds [56]. This leads to some bright objects with high reflectance in visible bands cannot be excluded effectively, such as the roads and buildings in the coast and urban area scenes in Fig. 9. Consequently, an adaptive gating mechanism is proposed to select valid annotations instead of combining additional thermal bands or auxiliary data as in previous studies [16], [17], [18], [19]. Given the adaptive selection of valid cloud annotations as an example, the initial HOT annotation $\mathcal{A}_{\text{init}}$ is regarded as the baseline representing the most exact cloud annotation results. Then, the affinity between multithreshold cloud annotations $\{\mathcal{A}^c_{t_k} | k \in K\}$ and $\mathcal{A}_{\text{init}}$ is calculated by

$$\alpha_{t_k} = \frac{\langle \mathcal{A}^c_{t_k}, \mathcal{A}_{\text{init}} \rangle}{\|\mathcal{A}^c_{t_k}\| \, \|\mathcal{A}_{\text{init}}\|} \quad \forall k \tag{4}$$

where $\langle , \rangle$ and $\|\cdot\|$ denote the dot production operator and ℓ_2 norm, respectively. Subsequently, the gating mechanism

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utilizes learned affinity weights to determine the consistency between the current threshold annotation and its adjacent threshold annotation ($\mathcal{A}^c_{t_k}$ versus $\mathcal{A}^c_{t_{k+1}}$). If $\alpha_{t_k} = \alpha_{t_{k+1}}$, it indicates that $\mathcal{A}^c_{t_k}$ is the same as $\mathcal{A}^c_{t_{k+1}}$. In other words, the threshold t_k is large enough to annotate precise cloud pixels. The gating mechanism will drop t_{k+1} from $\{t_k | k \in K\}$; otherwise, it will be retained. As shown in the blue box in Fig. 4, thresholds t_2 and t_3 achieve the same cloud annotations and equal affinity weights ($\alpha_{t_2} = \alpha_{t_3} = 0.6472$). In this case, t_2 is sufficient to precisely annotate cloud pixels, and increasing the threshold further may even lead to all cloud pixels being ignored. For example, no cloud pixels are activated in $\mathcal{A}^c_{t_4}$. Hence, t_3 is discarded. Only $\mathcal{A}^c_{t_1}$ and $\mathcal{A}^c_{t_2}$ are utilized for pseudo labeling. The adaptive gating operation can be formalized by the adjustment step index as follows:

$$K' = \{k \in K \mid \alpha_{t_k} \neq \alpha_{t_{k+1}}\} \tag{5}$$

where K' stands for the effective steps of threshold adjustments. Meanwhile, the HOT thresholds for precise cloud annotations are updated accordingly $\{t_k | k \in K'\}$. The learned affinities are normalized by $\alpha_{t_k} = \exp(\alpha_{t_k}) / \sum_{k \in K'} \exp(\alpha_{t_k})$ for pseudo labeling. Finally, the pseudo cloud mask CM can be derived as

$$\mathrm{CM} = \sum_{k \in K'} \alpha_{t_k} \times \mathcal{A}^c_{t_k}. \tag{6}$$

The adaptive selection of valid boundary annotations is identical to the above. Fig. 4 also shows this procedure in the green box. When getting valid boundary annotations, the pseudo BM can be derived as

$$\mathrm{BM} = \mathcal{A}^b_{t_{K'}} - \sum_{k \in K'-1} \alpha_{t_k} \times \mathcal{A}^b_{t_k} \tag{7}$$

where $\mathcal{A}^b_{t_{K'}}$ represents the most complete annotation that contains all possible cloud pixels. Theoretically, CM, BM $\in [0, 1]$, where 1 means the absolute cloud and 0 denotes the clear background. Its value indicates the confidence that a pixel may belong to a cloud. These adaptively annotated pseudo masks are then treated as ground truths to optimize HCDNet.

It should be stated that the physical rule used in RAPL is not set in stone. Section V discusses the impact of different physical rules in RAPL on detection capability.

Algorithm 1 RAPL

Input:
remote sensing images: $\boldsymbol{I} = \{I^{(n)}\}|_1^N$,
inital HOT threshold: t_{init}.
Output:
pseudo cloud masks: $\boldsymbol{CM} = \{CM^{(n)}\}|_1^N$.
pseudo boundary masks: $\boldsymbol{BM} = \{BM^{(n)}\}|_1^N$.

```
for I^(n) ∈ I do
    Adjust the initial threshold bidirectionally for more precise cloud annotations and more complete boundary annotations according to Equ. 2.
    Annotate potential cloud and boundary pixels based on HOT according to Equ. 1 and Equ. 3.
    Calculate the affinity between the initial HOT annotation and both the multithreshold cloud annotations and boundary annotations separately according to Equ. 4.
    Select valid cloud annotations and boundary annotations separately by the adaptive gating mechanism according to Equ. 5.
    Output the pseudo cloud mask according to Equ. 6.
    Output the pseudo boundary mask according to Equ. 7.
end
return CM, BM
```

C. Deformable Boundary Refining Module

Clouds exhibit complex geometric structures and varying transparency, ranging from fiber to lump, which poses challenges for accurate boundary detection. To address this issue, skip connections [42], [44], [57], pooling pruning [27], and customized loss function [22] have been proposed to enhance spatial details. However, the standard CNN is inherently limited to model complex geometric transformations due to their fixed sampling locations in convolution kernels. Inspired by the deformable convolution [58], this work proposes the DBRM to enhance the modeling capability of spatial transformation and capture precise boundaries from translucent cloud regions.

Unlike DCNet [23], HCDNet introduces DBRM as an individual module instead of directly replacing standard convolution with deformable convolution. As shown in Fig. 5, DBRM is designed as a residual structure with a short connection. The inputs of DBRM are extracted multiscale features $\{X^{(i)}\}|_{i=1}^L$ of the encoder, and the outputs are refined spatial features $\{B^{(i)}\}|_{i=1}^L$ corresponding to inputs, where the superscript i denotes the layer index. Since shallow layers of the neural network comprise fine-grained contextual features [59], DBRMs are attached to the first three layers of the encoder to capture spatial details.

More concretely, DBRM incorporates two deformable convolutional blocks to model the spatial transformations of clouds by irregular sampling grids. Each block contains four deformable convolutional layers. Unlike the fixed sampling grid of standard convolution, the sampling locations in deformable convolutional layers are augmented with learnable offsets $\{\triangle p_m | m = 1, \ldots, M\}$, where M means the number of sampling locations. For example, the grid $\mathcal{R} = \{(-1, -1), (-1, 0), \ldots, (0, 1), (1, 1)\}$ defines the sampling locations of a 3×3 convolution kernel. In standard convolutions, for each location p_0 on the output feature map y, we have

$$y^{(i)}(p_0) = \sum_{p_n \in \mathcal{R}} W^{(i)}(p_n) \cdot X^{(i)}(p_0 + p_n) \tag{8}$$

where p_n enumerates the locations in the receptive field $\mathcal{R}$, while in deformable convolutions, (8) becomes

$$y^{(i)}(p_0) = \sum_{p_n \in \mathcal{R}} W^{(i)}(p_n) \cdot X^{(i)}(p_0 + p_n + \triangle p_m). \tag{9}$$

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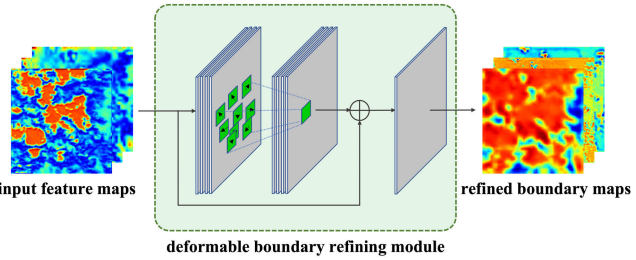


Fig. 5. Detailed structure of the proposed DBRM.

The sampling is on the irregular locations $p_n + \triangle p_m$, where $\triangle p_m$ are learned offsets from the spatial variation of clouds via additional convolutional layers. It enables irregular deformation of the sampling grid to model complex boundary structures. After that, the refined BMs can be obtained

$$B^{(i)} = \mathcal{F}(\mathcal{D}(X^{(i)}, W_i) + X^{(i)}) \quad \forall i \tag{10}$$

where the function $\mathcal{D}(\cdot)$ represents the deformable convolutional blocks and the function $\mathcal{F}(\cdot)$ means a 1×1 deformable convolutional layer for feature fusion. Finally, these refined BMs $\{B^{(i)}\}|_{i=1}^{L}$ are resized to the same resolution and then merged by an additional 1×1 standard convolutional layer to acclimate multiscale hidden spatial features and output boundary activation maps.

D. Optimization Objective

This work frames cloud detection as a semantic segmentation task to generate binary masks for clouds and backgrounds under a pixelwise classification paradigm. The optimization objective of HCDNet is defined as follows:

$$\mathcal{L} = \lambda \mathcal{L}_c + (1 - \lambda)\mathcal{L}_b + \mathcal{L}_h \tag{11}$$

where $\mathcal{L}_c$ and $\mathcal{L}_b$ denote the cloud detection loss and boundary prediction loss, respectively; $\mathcal{L}_h$ means the harmonic loss, which is devised to constrain a common semantic category between cloud and boundary activations while reserving spatial diversity; and λ is the balance factor, whose sensitivity analysis is given in Section IV-E3.

In the training phase, pseudo cloud masks annotated by RAPL are treated as ground truths to optimize the encoder and decoder of HCDNet. Following the independent manner of binary cross-entropy loss, the cloud detection loss $\mathcal{L}_c$ is formulated as:

$$\mathcal{L}_c = -\sum_{n=1}^{N} \left[\mathrm{CM}^{(n)} \log\left(P_c^{(n)}\right) + (1 - \mathrm{CM}^{(n)}) \log\left(1 - P_c^{(n)}\right)\right] \tag{12}$$

where $P_c^{(n)}$ means the cloud prediction and N represents the number of images in the dataset.

Correspondingly, HCDNet leverages BMs annotated by RAPL to optimize DBRM, enhancing the spatial perception of transparent cloud regions and activating sharp boundaries. Similar to (12), the boundary prediction loss $\mathcal{L}_b$ is calculated as follows:

$$\mathcal{L}_b = -\sum_{n=1}^{N} \left[\mathrm{BM}^{(n)} \log\left(P_b^{(n)}\right) + (1 - \mathrm{BM}^{(n)}) \log\left(1 - P_b^{(n)}\right)\right] \tag{13}$$

where $P_b^{(n)}$ means the boundary prediction obtained from boundary activations by additional 1×1 convolutional layer and $\mathrm{BM}^{(n)}$ is the annotated BM.

Because only red and blue bands are considered, the BM annotated by RAPL cannot adequately suppress bright surfaces. As a result, DBRM inevitably amplifies the interference of bright ground objects when perceiving more translucent boundaries and thin clouds. As shown in Fig. 9, boundary activation maps overestimate the bright roads and buildings in the middle of the coast and urban area scenes. Different from traditional methods utilizing additional thermal bands or auxiliary tasks for better annotations [17], [19], [36], this work develops a harmonic loss $\mathcal{L}_h$ to improve the perception of clouds with nonuniform spectral reflectance and suppress the interference of bright backgrounds. Specifically, $\mathcal{L}_h$ simultaneously constrains cloud and boundary activations from the semantic and spatial perspectives. In the semantic space, the Wasserstein distance [60], [61] is employed to measure the similarity between cloud and boundary activations to constrain a common semantic representation. In the spatial space, we diversify the cloud and boundary activations by maximizing the standard deviation [62] to activate translucent boundaries and thin clouds with the same semantic category. $\mathcal{L}_h$ is formulated as follows:

$$\mathcal{L}_h = \sum_{n=1}^{N} \left[\beta \left\| P_c^{(n)} - P_b^{(n)} \right\|_w - (1 - \beta)\left(\delta_{P_c^{(n)}} + \delta_{P_b^{(n)}}\right)\right] \tag{14}$$

where $\|\cdot\|_w$ and $\delta_{(\cdot)}$ mean calculating the Wasserstein distance and standard deviation, respectively, and β is a balance factor.

IV. Experiment and Analysis

This section evaluates the proposed framework on three public RS image cloud detection datasets. Note that even though most of these datasets provide pixelwise labels annotated by experts, we only use them for evaluation but not for HCDNet training in any way.

A. Datasets

1) GF-1 Cloud and Cloud Shadow Cover Validation Data: The GF-1 dataset [19] includes 108 wide field-of-view level-2A scenes acquired from the GaoFen-1 satellite, covering various land types, such as the forest, barren, ice/snow, water, wetlands, and urban area. These selected images are produced after relative radiometric correction and systematic geometric correction. The approximate size of each image is $15\,700 \times 16\,200$ pixels, and the number of spectral bands is four. In addition, the reference masks for the quantitative evaluation are obtained manually by experienced experts, which contain cloud, cloud shadow, clear, and nonvalue. According to the common practice [21], [23], this experiment randomly selects images with a ratio of 40:40:28 for training, validation,

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and testing. The cloud shadow is also redefined as clear. All original images are sliced into patches of 320 × 320 pixels at the step of 60 pixels by sliding windows from left to right and top to bottom to meet limited computing resources. The strategy of overlapping division is adopted to ensure the local consistency of patches. In the testing phase, all patches are spliced to their original size for quantitative and qualitative evaluation. This dataset is available online.[1]

2) Landsat-8 Cloud Cover Assessment Validation Data: The L8 Biome dataset [33] contains 96 multispectral images captured by the Landsat-8 satellite operational land imager and thermal infrared sensor with a spatial resolution of 30 m. The approximate size of each image is 7500 × 7800 pixels, and the number of spectral bands is 11. These images cover eight kinds of natural scenes, including barren, forest, grass/crop, shrubland, snow/ice, urban, water, and wetlands. Each image contains rich multispectral bands, such as the near-infrared band, the shortwave infrared band, and the thermal infrared band, which prove favorable for cloud extraction in further experiments. Furthermore, the authors manually annotated five categories: cloud, thin cloud, clear, cloud shadow, and nonvalue. In this experiment, the ratio of training, validation, and testing sets is 66:10:20 images. The images in each set are selected randomly. All original images are also sliced into 320 × 320 pixels at the step of 60 pixels for training and then stitched back for evaluation. For reasonable assessment, the thin cloud category is redefined as the cloud, and the cloud shadow category is redefined as the clear. Apart from the panchromatic band with a different spatial resolution, this work takes all spectral bands as the input to the network. This dataset is available online.[2]

3) WDCD Data: The WDCD dataset [27] is a patch annotated RS image dataset collected from the GaoFen-1 satellite. The training set consists of 206 384 patches with binary labels, cropped from 622 original images. They cover different land types, such as farmland, wasteland, coastal, and mountain. Each patch has a size of 250 × 250 pixels, and the number of spectral bands is four. With the aid of visual interpretation, domain experts labeled the patch without any cloud pixels as the negative sample and annotate the patch with a cloud cover rate over 25% as the positive sample for training stability. In addition, the authors manually annotated 24 cropped GaoFen-1 images with pixelwise reference masks to evaluate the cloud detection performance. The approximate size of each image is 4500 × 4500 pixels, and the number of spectral bands is four. In this experiment, we randomly select eight images for validation and 16 images for testing. Then, all validation and testing images are divided into patches of 250 × 250 pixels at a step size of 60 pixels. The final result of the whole image is evaluated by stitching together each patch result. This dataset is available online.[3]

B. Evaluation Metrics

We use the OA, F1 score (F1), mean intersection over union (mIoU), and kappa coefficient (Ka) as quantitative metrics to evaluate the performance of HCDNet. Here, all quantitative scores are calculated using the pixelwise ground truths provided by the dataset as references. Note that these pixelwise ground truths only participate in quantitative evaluations and are forbidden in model training.

These quantitative metrics are defined as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}+\mathrm{TN}} \tag{15}$$

$$\mathrm{F1} = \frac{2\times\mathrm{TP}}{2\times\mathrm{TP}+\mathrm{FN}+\mathrm{FP}} \tag{16}$$

$$\mathrm{mIoU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}} \tag{17}$$

$$\mathrm{Ka} = \frac{p_0-p_e}{1-p_e} \tag{18}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively. Besides, $p_0 = ((\mathrm{TP}+\mathrm{TN})/(P+N))$ and $p_e = ((P(\mathrm{TP}+\mathrm{FP})+N(\mathrm{TN}+\mathrm{FN}))/((P+N)^2))$, where P and N denote the number of cloud and clear sky pixels in ground truths, respectively.

To assess the efficiency of compared methods, we evaluate the number of parameters of the model (denoted as Params) and the number of floating-point operations (FLOPs) for the model to compute a sliced patch with 320 × 320 pixels. In addition, the inference time of each WDCD model is also evaluated at the whole test set as well as at an image patch with 320 × 320 pixels cropped from the GF-1 dataset.

C. Implementation Details

This work uses the ResNet-34 [53] as the encoder to extract semantic features. However, the repeated combination of max pooling and striding at consecutive layers is not ideal for cloud detection since detailed spatial information is decimated. Thus, an atrous convolution [54] with rates determined by the desired output_stride = 8 is implemented to maintain the spatial resolution. Output_stride defines the ratio of input image resolution to encoder output resolution. We also replace the original 7 × 7 × 3 input layer of ResNet-34 with 7 × 7 × c convolutional layer to fit the multispectral image, where c is the number of spectral bands of input images. Each convolutional layer in HCDNet is followed by a batch normalization layer and a rectified linear unit (ReLU) activation layer.

In the GF-1 and WDCD datasets, the initial threshold is set to 0.13 according to the MFC recommendation [19], and the step size $s = 0.02$. Similarly, the initial threshold on the L8 Biome dataset is set to 0.8 according to the Fmask description [16], and the step size $s = 0.2$. The final segmentation threshold ξ is defined as 0.4 for all three datasets.

The proposed HCDNet is implemented with the PyTorch framework. The operating system is Ubuntu 16.04 equipped with two NVIDIA Tesla-P100 16G GPUs. The HCDNet is trained for ten epochs with a batch size of four patches. The stochastic gradient descent (SGD) algorithm is used as an optimizer for HCDNet training with a momentum of 0.9. A "poly" learning rate update strategy is chosen with an initial learning rate of 1×10^{-4} and a power of 0.9. The compared classic

[1] http://sendimage.whu.edu.cn/en/mfc-validation-data
[2] https://landsat.usgs.gov/landsat-8-cloud-cover-assessment-validation-data
[3] https://github.com/weichenrs/WDCD

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methods are trained with the same experimental settings. Only one GPU is used in the test phase.

D. Comparison With State-of-the-Arts

We compare the proposed HCDNet with several state-of-the-art cloud detection methods, including physics-based algorithms and CNN-based methods that are fully or weakly supervised. These compared approaches are summarized as follows.

1) *Traditional Physics-Based Methods:* MFC [19] and Fmask [16] use physical rules based on cloud properties to separate cloud and clear pixels.
2) *Fully Supervised Methods:* DCNet [23] and DABNet [22] are CNN-based cloud detection methods using pixelwise supervision. They utilize deformable convolution layers to improve the model capability of spatial transformation, facilitating precise boundaries in detection results. The reference implementations in PyTorch have been published online.[4]
3) *Weakly Supervised Methods:* WDCD [27], Matting [29], and WSFNet [28] utilize patch labels indicating only the presence or absence of cloud in cropped RS patches for cloud detection. Similarly, AuxSegNet [63] is a weakly supervised multitask deep network in natural images, which leverages saliency detection and image classification as auxiliary tasks to learn more discriminative features for segmentation. We integrate it into comparisons to explore whether the semantic segmentation model in natural scenes is appropriate for cloud detection in RS images. In addition, scExtractor [49] utilizes centerline scribbles for road extraction in RS images. DeepCloud [32] erodes the Fmask predictions as scribble annotations for pixelwise cloud detection. Their reference implementations in PyTorch have been published online.[5]

1) Experimental Results of GF-1 Dataset: We first compare the HCDNet with other prevalent methods on the GF-1 dataset. Table I reports quantitative results, where "Supervision" means the supervision styles used in model training, including pixelwise (pixel), patch, scribble, and rule. Among them, the rule denotes using pseudo cloud mask and BM annotated by RAPL as weak supervision.

To be specific, MFC is an established physical method based on cloud properties and expert knowledge. Therefore, MFC does not require additional supervision and should be compared with fully supervised cloud detection models [21], [26]. In general, the performance of CNN-based cloud detection methods is better than that of MFC. This is because MFC only considers local spectral variations and geometric textures, while CNN-based methods simultaneously characterize the global and local features of clouds. When using pixel supervision, HCDNet outperforms DABNet and DCNet but has more parameters. This is mainly attributed to the incorporation of DBRM and harmonic loss to learn more representation features from weak supervision. Moreover, since the DBRM is plugged into shallow layers as individual modules instead of directly replacing the standard convolutional layers, the FLOPs of HCDNet are less than those of DCNet.

In terms of WDCD, HCDNet achieves excellent performance with reasonable Params and FLOPs. DeepCloud ranks second in OA among the compared methods by using MFC outputs as weak supervision. As shown in Fig. 6, DeepCloud can identify opaque clouds distinctly but fails to suppress backgrounds around cloud edges. Similarly, scExtractor also amplifies interference from background regions. These results suggest that the superiority of DeepCloud and scExtractor mainly stems from the precise annotation of MFC. They are sensitive to texture features but lack sufficient semantic information for cloud recognition, leading to commission errors of background. Since patch labels only provide global semantic categories, WDCD, Matting, WSFNet, and AuxSegNet pursue regional accuracy while neglecting the spatial details of clouds. As shown in Fig. 6, the bright rivers and lakes are falsely alarmed as clouds in the wetland scene, and WSFNet predicts several mosaic blurs in the forest scene when clouds purely cover a patch. In particular, AuxSegNet leverages saliency detection as an auxiliary task to learn more spatial features, which obtains more accurate detection results than cognate methods at the expense of multiplied Params and FLOPs. These results prove that spatial features play a significant role in cloud detection. Benefiting from the comprehensive consideration of the spectral and spatial characteristics of clouds, HCDNet can detect clouds with different transparency and suppress bright backgrounds. As shown in Fig. 6, only a small portion of thin clouds blended with backgrounds are neglected in the forest scene, and bright ground objects are filtered effectively in the wetland scene.

2) Experimental Results of L8 Biome Dataset: Compared with the GF-1 dataset, the L8 Biome dataset contains more fine-grained spectral bands. HCDNet takes all spectral bands as input for a reasonable comparison, except for the panchromatic band with a different spatial resolution.

Table I and Fig. 7 show the quantitative and qualitative results, respectively. Fmask is a reliable physical cloud detection method for Landsat-8 images, which exhibits high OA on the L8 Biome dataset due to its well-designed physical rules. Like the GF-1 dataset, HCDNet* also outperforms DABNet and DCNet and boosts the Fmask performance from 89.40% to 97.27% in terms of F1.

For WDCD, HCDNet is consistently superior compared to other methods. Because the L8 Biome dataset contains more complex scenes with thin clouds, the weakness in recovering spatial details becomes more prominent among weakly supervised models based on patch supervision. As shown in Fig. 7, WDCD, Matting, and WSFNet face a dilemma of category ambiguity caused by blurred spectra, making translucent boundaries and thin clouds less detectable. AuxSegNet still exhibits better detection results for thin clouds and translucent boundaries compared to cognate methods with the help of additional saliency detection. Despite using identical scribble supervision, scExtractor and DeepCloud give inconsistent predictions. DeepCloud employs a dual branch to extract contextual and spectral features to detect cloud

[4]https://github.com/NiAn-creator/deformableCloudDetection
[5]https://github.com/NiAn-creator/weakCloudDetectionMethods

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TABLE I
COMPARISON OF HCDNET WITH OTHER METHODS ON THE GF-1 AND L8 BIOME TEST SET

Methods	Supervision	Performance on GF-1 Test Set				Performance on L8 Biome Test Set				Efficiency	
		OA(%)	F1(%)	mIoU(%)	Ka(%)	OA(%)	F1(%)	mIoU(%)	Ka(%)	Params(MB)	FLOPs(G)
MFC [19]	/	94.42	91.30	88.06	87.20	/	/	/	/	/	/
Fmask [16]	/	/	/	/	/	92.27	89.40	84.69	83.39	/	/
DABNet [22]	*pixel*	96.60	94.39	92.30	91.95	95.16	96.49	89.36	88.68	4.12	16.58
DCNet [23]	*pixel*	96.86	94.97	92.97	92.68	94.58	96.07	88.22	87.38	7.12	32.42
HCDNet*	***pixel***	**96.97**	**95.13**	**93.21**	**92.94**	**96.27**	**97.27**	**91.80**	**91.40**	37.14	27.66
WDCD [27]	*patch*	91.65	83.74	80.70	78.14	87.40	87.87	77.60	74.92	10.86	939.17
Matting [29]	*patch*	92.40	86.24	81.02	78.97	81.27	87.32	62.39	52.38	50.61	136.82
WSFNet [28]	*patch*	90.39	76.35	76.54	72.06	84.59	85.47	73.22	69.36	45.05	26.79
AuxSegNet [63]	*patch*	93.97	90.73	87.23	86.27	92.63	89.17	84.94	83.60	115.25	105.38
scExtractor [49]	*scribble*	90.85	84.72	80.62	78.21	86.87	80.61	74.73	70.68	31.32	17.98
DeepCloud [32]	*scribble*	95.26	92.48	89.66	89.02	92.56	88.97	84.75	83.36	1.89	10.47
HCDNet	***rule***	**96.56**	**92.75**	**91.03**	**90.49**	**93.83**	**91.05**	**87.29**	**86.35**	37.14	27.66

: HCDNet is fully supervised model using pixel-wise ground truths.

pixels as entirely as possible, resulting in false alarms of backgrounds. In contrast, scExtractor does not consider the spectral characteristics of clouds and therefore pays more attention to opaque cloud regions, resulting in omission errors of translucent clouds. As for HCDNet, it combines a harmonic loss to balance the semantic identification and spatial activation, enabling accurate detection of clouds with nonuniform spectral reflectance and predicting sharp boundaries. Section IV-E3 specifies the effectiveness of the harmonic loss.

3) Experimental Results of WDCD Dataset: We further evaluate the performance of HCDNet with competing approaches on the WDCD dataset. As the training set of the WDCD dataset lacks pixelwise annotations, only weakly supervised methods are compared. Table II shows that HCDNet is more effective than compared strategies.

The images in the WDCD test set are cropped from the GaoFen-1 satellite images, approximately one-tenth the size of the original images. It allows a more detailed assessment of the compared methods under different land surfaces and various cloud types. As shown in Fig. 8, the wasteland scene comprises bright bare lands and translucent clouds. Under such circumstances, the spectral reflectance of clouds is blended with backgrounds, which brings significant challenges for cloud detection. Consequently, these WDCD models have exhibited similar omission errors of thin clouds. Especially for methods that rely on patch supervision, they aim to learn a high-level semantic concept of clouds from global guidance rather than precise pixel-level spatial information. This is particularly problematic for methods such as WDCD, Matting, WSFNet, and AuxSegNet, which are prone to misclassifying bright pixels as clouds and ignoring translucent cloud regions. It makes the model prediction not align with the actual cloud contours. Under scribble supervision, DeepCloud still tends to activate all potential cloud pixels, resulting in overestimation around boundaries. The scExtractor lags behind DeepCloud as it fails to consider the spectral properties of clouds. The proposed HCDNet achieves comparable performance to DeepCloud in wasteland scenes, as both struggles to activate extremely translucent cloud pixels. The difference is that DeepCloud is more progressive to visual features in the image and tends to activate more potential cloud pixels. HCDNet is more conservative to cloud identification and tends to detect cloud pixels precisely to achieve higher precision.

TABLE II
COMPARISON OF HCDNET WITH OTHER METHODS ON THE WDCD TEST SET

Methods	Supervision	OA(%)	F1(%)	mIoU(%)	Ka(%)
WDCD [27]	*patch*	94.81	83.82	83.07	80.73
Matting [29]	*patch*	93.06	79.02	79.57	75.42
WSFNet [28]	*patch*	92.00	79.83	78.64	74.85
AuxSegNet [63]	*patch*	92.54	76.90	76.37	72.33
scExtractor [49]	*scribble*	92.94	76.90	76.37	72.33
DeepCloud [32]	*scribble*	95.95	86.43	85.72	84.05
HCDNet	***rule***	**96.70**	**88.83**	**88.06**	**86.90**

E. Ablation Studies

1) Effectiveness of RAPL: Table III assesses the effectiveness of the proposed RAPL on the GF-1 validation set. Here, HOT stands for the spectral reflectance rule with an initial threshold based on HOT. In this experiment, the initial threshold is 0.13 according to the MFC recommendation. As visualized in Fig. 9, the HOT initial threshold only identifies opaque clouds, failing to detect translucent boundaries and thin clouds in the mountain and wetland scenes. Conversely, the HOT initial threshold overestimates bright roads and buildings as clouds in the coast and urban area scenes. It is evident that the HOT initial threshold strives for globally optimal performance, leading to local errors. To tackle this problem, RAPL performs bidirectional threshold segmentation to annotate pixels more likely to be clouds or backgrounds for HCDNet training. Specifically, HOT *w/* RAPL^c refers to only increasing the initial threshold for cloud masks labeling, whereas HOT *w/* RAPL^b refers to only decreasing the initial threshold for BMs labeling. As shown in Fig. 9, RAPL cloud masks comprise more explicit semantic categories, and RAPL BMs contain more complete spatial contours.

On this basis, HCDNet regards these pseudo masks annotated by RAPL as ground truths to supervise network training. As assessed in Table III, RAPL^c and RAPL^b bring 5.74% and

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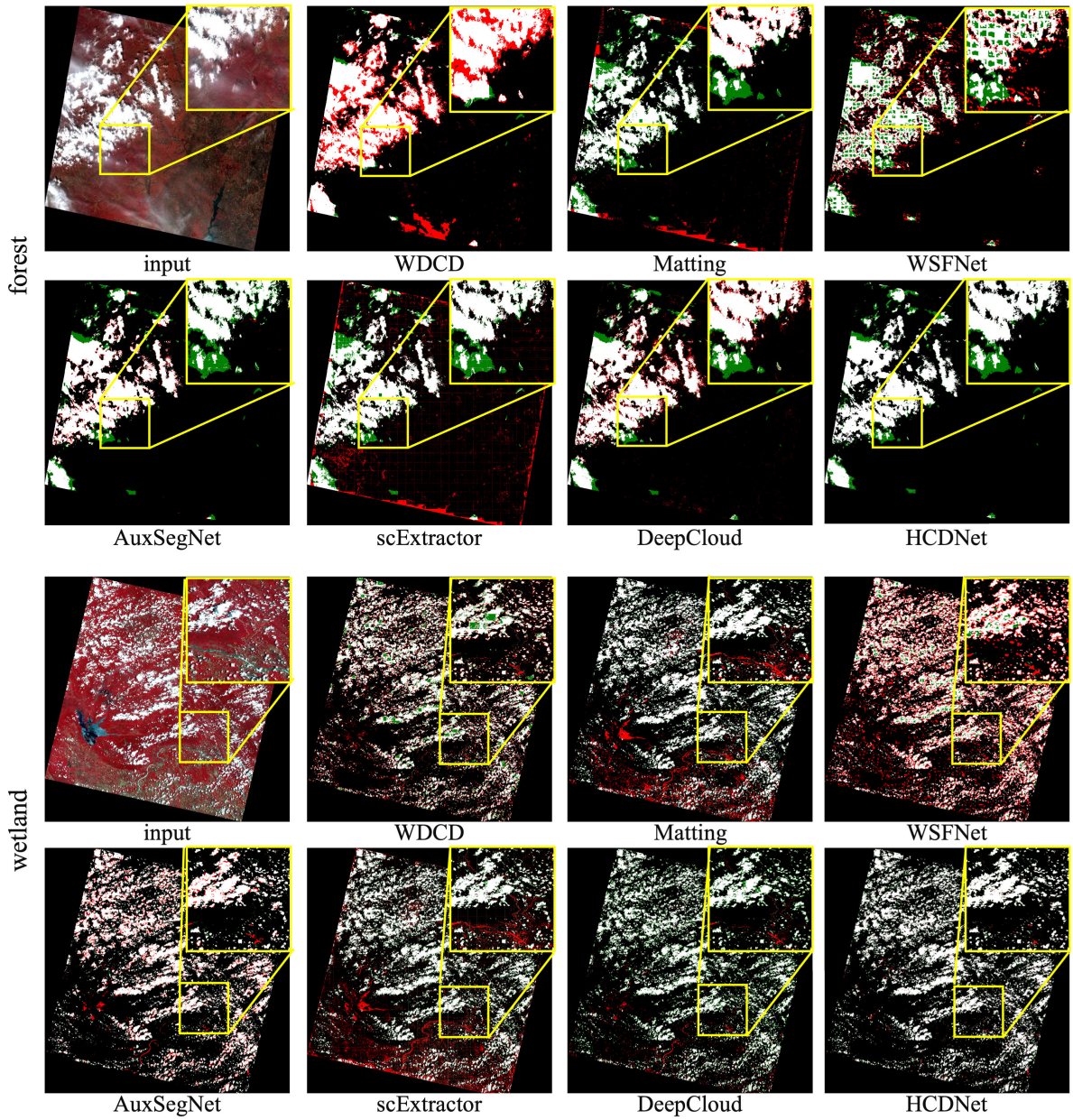


Fig. 6. Qualitative comparisons of cloud detection results on the GF-1 dataset. Black represents clear skies, white means clouds, red indicates clear skies that are falsely alarmed as clouds, and green indicates clouds that are omitted as clear skies. The yellow box highlights a local area for detailed comparison. Note that nonvalue pixels are also visualized as black colors but are not involved in the quantitative and qualitative evaluation.

TABLE III

EFFECTIVENESS OF THE RAPL ON THE GF-1 VALIDATION SET

Methods	OA(%)	F1(%)	mIoU(%)	Ka(%)
HOT	91.92	80.49	78.82	75.41
HOT *w/* RAPL^{c}	89.49	84.82	79.39	76.93
HOT *w/* RAPL^{b}	90.62	85.84	80.76	78.60
HOT *w/* HCDNet	89.97	83.96	79.38	76.66
HOT *w/* RAPL^{c}+HCDNet	94.09	89.70	86.69	85.60
HOT *w/* RAPL^{b}+HCDNet	93.87	90.22	86.82	85.75
HOT *w/* RAPL+HCDNet	**94.52**	**91.11**	**88.03**	**87.16**

TABLE IV

EFFECTIVENESS OF THE DBRM ON THE GF-1 VALIDATION SET

Methods	OA(%)	F1(%)	mIoU(%)	Ka(%)
VGG-16 [64]	91.51	87.37	82.78	81.04
ResNet-50 [53]	91.91	85.16	81.81	79.73
ResNet-34 [53]	93.09	87.83	84.54	83.05
ResNet-34 *w/* BRM	94.29	90.03	87.10	86.05
ResNet-34 *w/* DBRM	**94.52**	**91.11**	**88.03**	**87.16**

6.26% improvements in terms of F1 over directly using the initial threshold annotation as supervision (HOT *w/* HCDNet), respectively. Moreover, HOT *w/* RAPL + HCDNet further boosts the detection accuracy by using cloud mask and BM as supervision together. These results indicate that the superior performance of HCDNet mainly stems from its ability to learn more accurate semantic and spatial features about clouds at

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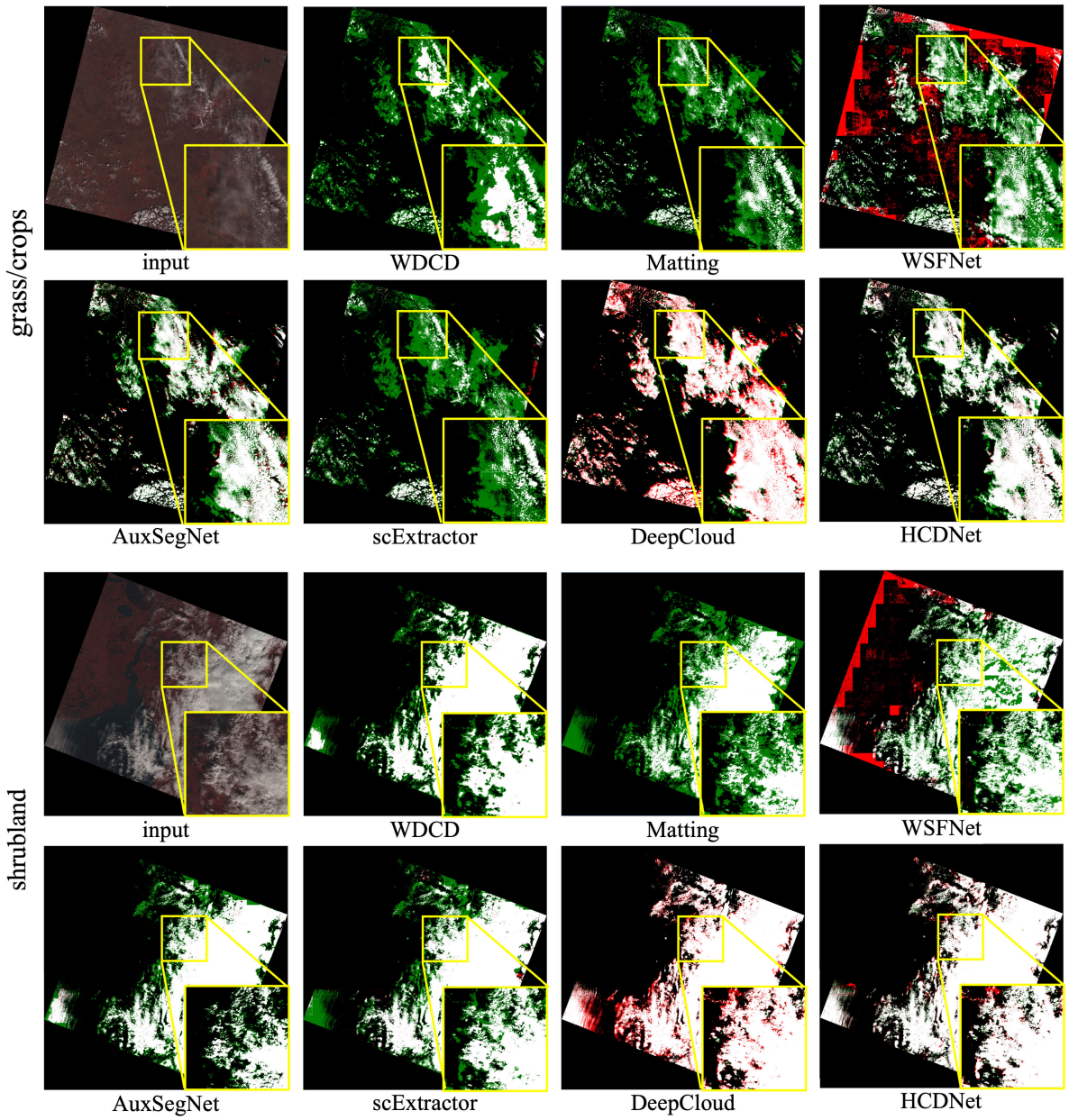


Fig. 7. Qualitative comparisons of cloud detection results on the L8 Biome dataset. Black represents clear skies, white means clouds, red indicates clear skies that are falsely alarmed as clouds, and green indicates clouds that are omitted as clear skies. The yellow box highlights a local area for detailed comparison. Note that nonvalue pixels are visualized as black but are not involved in the quantitative and qualitative evaluation.

TABLE V
EFFECTIVENESS OF THE HARMONIC LOSS ON THE GF-1 VALIDATION SET

Methods	OA(%)	F1(%)	mIoU(%)	Ka(%)
HCDNet *w/o* $\mathcal{L}_h$	94.27	90.32	87.28	86.25
HCDNet	**94.52**	**91.11**	**88.03**	**87.16**

the same time rather than simply extracting additional spatial features from BMs.

Although RPAL can enhance the confidence of pixels more likely to be clouds through adaptive thresholding, which guides HCDNet to focus on these features during training. It cannot exactly exclude bright backgrounds or activate translucent clouds. Therefore, the pseudo mask generated by RPAL inevitably contains false annotations, such as the omitted translucent clouds and overestimated bright buildings in Fig. 9. Hence, a harmonic loss is designed to mitigate the deception of false labels during training. More detailed analyses are given in Section IV-E3.

2) Effectiveness of DBRM: We execute DBRMs on shallow layers of the encoder to capture more detailed spatial features. To verify the effectiveness of DBRM, we conduct ablation studies with different settings in Table IV.

First, we assess the impact of different encoders. As shown in Table IV, ResNet-34 achieves comparable accuracy to VGG-16 [64]. When a deeper ResNet-50 is adopted, HCDNet performance degrades. It shows that the repeated

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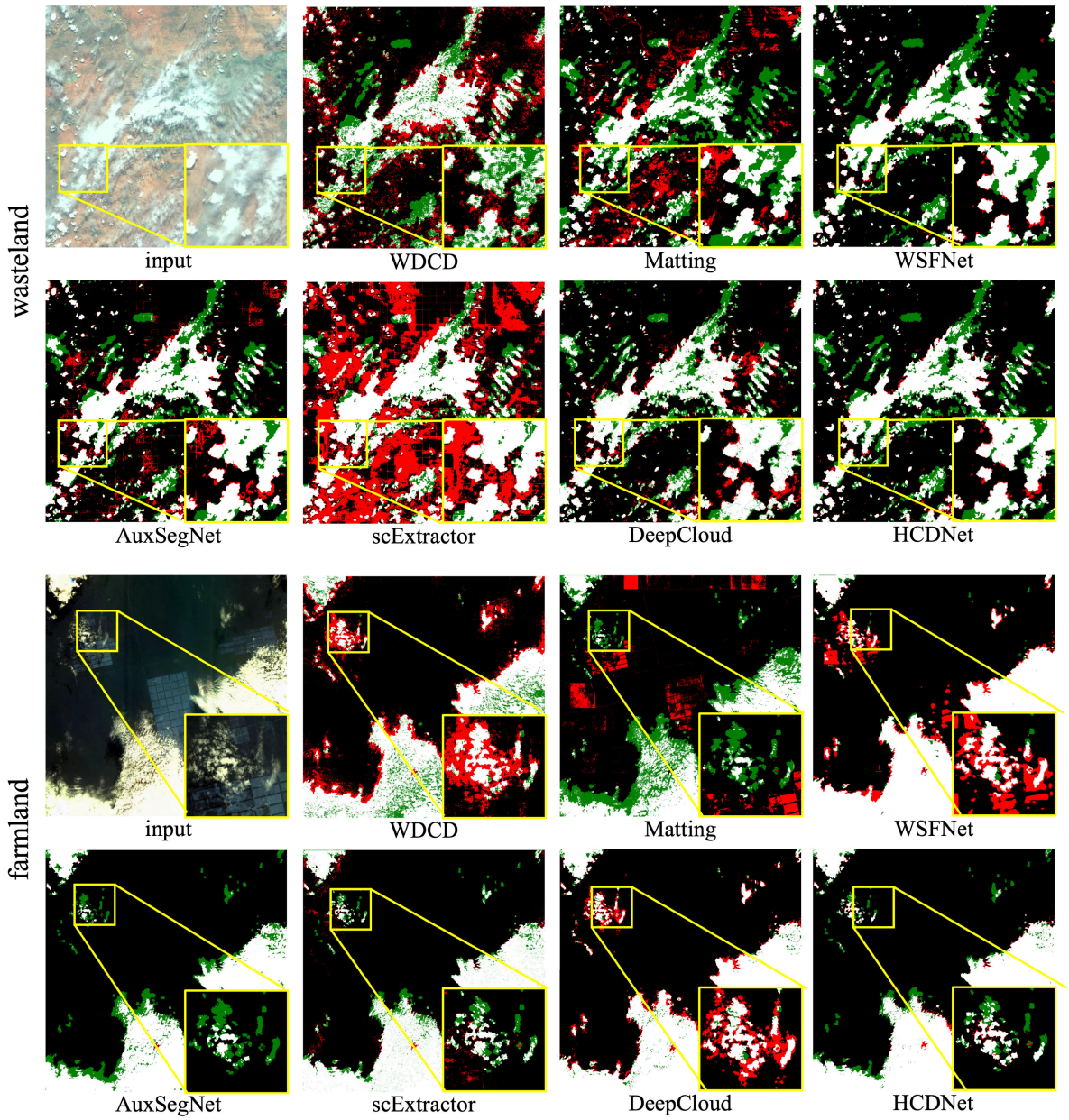


Fig. 8. Qualitative comparisons of cloud detection results on the WDCD dataset. Black represents clear skies, white means clouds, red indicates clear skies that are falsely alarmed as clouds, and green indicates clouds that are omitted as clear skies. The yellow box highlights a local area for detailed comparison.

downsampling operations at deeper consecutive layers are not ideal for cloud detection since detailed spatial information is broken. Therefore, HCDNet stands on ResNet-34 as a baseline to evaluate the effectiveness of DBRM on the GF-1 validation set.

Subsequently, we verify the effectiveness of the proposed DBRM. In order to strengthen the contrast, the standard convolution layers are introduced to implement the regular boundary refining module (BRM), termed ResNet-34 *w/* BRM, which outperforms ResNet-34 by 2.20% F1 and 2.56% mIoU. After this, the deformable operation is incorporated to refine the irregular boundaries. As presented in Table IV, ResNet-34 *w/* DBRM further promotes the detection accuracy by 1.08% and 0.93% increases in terms of F1 and mIoU, respectively.

Fig. 7 presents several qualitative comparisons of DBRM. Compared to cloud activation maps, boundary activation maps are sensitive to spatial details and emphasize more translucent boundaries and thin clouds. These results declare that DBRM can characterize complex geometric structures and activate precise boundaries from among translucent cloud regions.

3) Effectiveness of Harmonic Loss: The harmonic loss devotes itself to recognizing clouds with nonuniform spectral reflectance and excluding bright surfaces. As shown in Table V, the harmonic loss brings 0.79% and 0.75% advances on F1 and mIoU for HCDNet, respectively.

Furthermore, we conducted the sensitivity analyses of the loss weight λ and β. As shown in Fig. 10, a higher β will cause HCDNet to overemphasize the semantic consistency between cloud and boundary activations, which will

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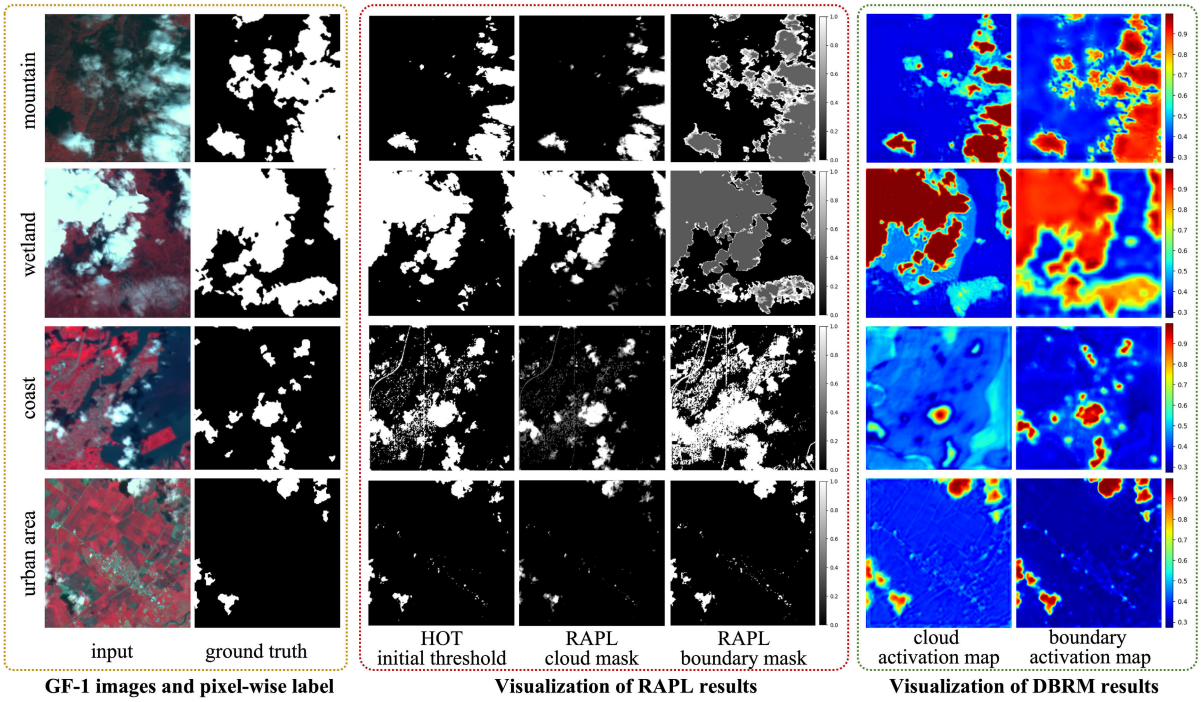


Fig. 9. Qualitative comparisons of the proposed RAPL and DBRM.

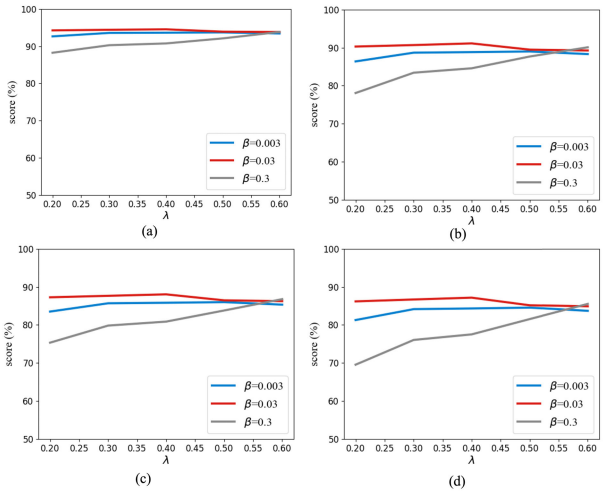


Fig. 10. Performance of HCDNet with different loss weights on the GF-1 validation set. (a) OA. (b) F1. (c) mIoU. (d) Ka.

TABLE VI
INFERENCE TIME OF COMPARED METHODS

Methods	GF-1(s)	L8 Biome(s)	WDCD(s)	patch*(s)
MFC [19]	3587	/	2959	/
Fmask [16]	/	3933	/	/
DABNet [22]	2422	1752	/	0.24
DCNet [23]	1150	832	/	0.10
WDCD [27]	7200	5208	5526	0.61
Matting [29]	3925	2839	3012	0.33
WSFNet [28]	5352	3872	4107	0.45
AuxSegNet [63]	2372	1716	1821	0.20
scExtractor [49]	1008	729	774	0.09
DeepCloud [32]	998	722	766	0.08
HCDNet	**850**	**653**	**652**	**0.07**

*: Patch is croped from GF-1 dataset with the size of 320 × 320 and four spectral bands. "s" denotes the time cost in seconds.

damage the spatial diversity. In general, the harmonic loss is impervious to β changes. As far as cloud detection and boundary prediction are concerned, HCDNet is not sensitive to the variation of segmentation weight λ. Experimental results prove that the proposed HCDNet can learn adequate feature representation from numerous weak annotations without using extra training tricks. Instead of fine-tuning the loss weight more exhaustively, this work chooses the relatively stable $\lambda = 0.4$ and $\beta = 0.03$ as the best experimental settings.

F. Time Efficiency

Table VI evaluates the time efficiency of compared cloud detection methods. In the implementation, we test the inference time on the entire test set of GF-1, L8 Biome, and WDCD datasets. The traditional MFC for the GF-1 dataset and Fmask for the L8 Biome dataset are executed in the CPU mode using an Intel Core i7-8700. The CNN-based methods are tested with the NVIDIA Tesla-P100 GPU. As these methods perform inference based on patches and then stitch them back to the original size as the final result for evaluation, the inference time of patches on the GF-1 test set is also assessed for these methods.

Generally, CNN-based methods are computationally more efficient than traditional MFC and Fmask due to the use of GPU. However, it is exceptional for WDCD and Matting since they need to compute hidden features with larger spatial resolutions or use auxiliary tasks to recover spatial details during the testing phase, leading to longer inference time.

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TABLE VII
SENSITIVITY ANALYSIS OF DIFFERENT PSEUDO LABELING STRATEGIES ON THE GF-1 VALIDATION SET

Methods	Ablation	OA(%)	F1(%)	mIoU(%)	Ka(%)
OTSU	——	82.64	72.36	67.11	59.71
	HCDNet	89.45	79.55	76.39	72.81
Red	RAPL	89.20	81.36	77.65	74.57
	RAPL+HCDNet	94.35	90.65	87.56	86.59
VBR	RAPL	86.49	79.64	74.10	69.96
	RAPL+HCDNet	90.50	83.63	79.73	76.90
HOT	RAPL	89.49	84.82	79.39	76.93
	RAPL+HCDNet	94.52	**91.11**	88.03	87.16
MFC	RAPL	94.11	90.75	87.38	86.42
	RAPL+HCDNet	**94.58**	91.09	**88.08**	**87.20**

TABLE VIII
EFFICIENCY OF DIFFERENT PSEUDO LABELING STRATEGIES ON THE GF-1 TRAINING SET

Methods	Accuracy of Pseudo Annotations				Efficiency	
	OA(%)	F1(%)	mIoU(%)	Ka(%)	Time(s)	Mem(MB)
HOT	91.18	80.49	78.28	74.81	917	1999.3
MFC	94.70	88.67	86.48	85.22	4696	17120
HOT *w/*RAPL	92.31	83.82	81.27	78.78	1057	3552.2
MFC *w/*RAPL	94.87	88.99	86.84	85.65	5477	17731.3

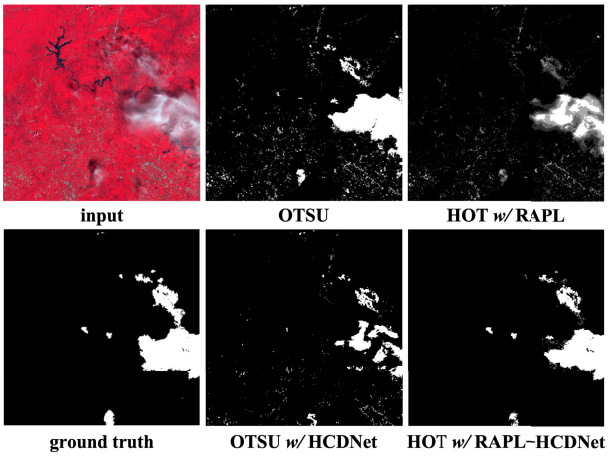


Fig. 11. Qualitative comparisons of HCDNet with different pseudo labeling strategies on the GF-1 validation set. HOT *w/* RAPL represents the cloud mask annotated by RAPL with the HOT rule.

The inference time of DeepCloud and scExtractor is almost the same as HCDNet, but their cloud detection accuracy is inferior, as discussed in Section IV-D. Although DeepCloud has fewer Params and FLOPs, it is slower than HCDNet due to its higher spatial resolutions in hidden layers. Among compared methods, HCDNet achieves satisfactory accuracy with the shortest inference time during testing.

V. DISCUSSION

This work devises an RAPL algorithm to annotate pseudo cloud masks and BMs as weak supervision adaptively according to cloud spectral properties without manual intervention. Empirically, the smaller the gap between pseudo masks and pixelwise ground truths, the more reliable features can be learned by the model for better performance. In the following, we specify whether annotation results will affect the detection capability and how.

First, we verify whether different pseudo labeling strategies affect the detection capability. This work compares the impact of two strategies to verify this question: the automatic thresholding method and the physical rule method. Regarding the automatic thresholding segmentation for images, OTSU [65] is widely used to find the best threshold from a gray-level statistical histogram to segment the gray image into a binary image. It has also been applied to cloud detection tasks [66], [67]. For example, Kreuter et al. [66] used OTSU thresholding on the difference between red and blue bands to detect clouds. Xiang [67] proposed an OTSU-based k-means algorithm to achieve automatic cloud detection. Inspired by this, we utilize OTSU to segment clouds and treat its results as weak supervision to optimize HCDNet. As shown in Fig. 11, OTSU only activates objects with bright colors. OTSU *w/* HCDNet gives sparse cloud activations, resulting in the interference of cloud-like objects in the background and the omission of translucent clouds. Albeit simple in threshold selection, OTSU ignores the specific physical properties of clouds. On this basis, we devise the RAPL to annotate potential cloud regions, which is a physical rule method based on HOT. As shown in Fig. 11, HOT *w/* RAPL activates clouds with nonuniform spectral reflectance from backgrounds in the cloud masks, whose value explicitly presents the confidence of the pixel about the cloud. As reported in Table VII, HOT *w/* RAPL + HCDNet outperforms OTSU *w/* HCDNet by 11.56% F1 and 11.64% mIoU due to the explicit semantic and spatial guidance provided by RAPL. These results reveal that leveraging physical rules for pseudo labeling is beneficial to generating more reliable pseudo annotations and promoting cloud detection accuracy.

Subsequently, we verify whether different physical rules in RAPL affect the detection capability. In detail, we conduct experiments on four variations of physical rules to generate pseudo masks as weak supervision. First, we find that clouds always have higher reflectivity than clear sky pixels in the red band. In this case, a red band threshold is set to segment clouds [15]. Second, the ratio of the minimal and maximal reflectance in visible bands can be used to exclude bright backgrounds with salient color features from the extracted cloud masks [19]. Hence, a physical rule based on visible band ratio (VBR) is produced. The VBR is close to one when the pixel is a cloud. Third, because the spectral response of clouds between the blue and red wavelengths differs from that of most land surfaces [34], [35], the HOT is applied to separate clouds and clear sky pixels. Fourth, MFC is a classic physical algorithm that utilizes combined spectral rules and geometric features for cloud detection.

Table VII evaluates the accuracy of pseudo cloud masks annotated by RAPL. The proposed RAPL maintains a robust performance among various physical rules. It is mainly attributed to RAPL, which can annotate clouds with nonuniform spectral reflectance by adjusted thresholds and exclude unreasonable annotations by the adaptive gating mechanism. This prevents excessive activation of the background

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or suppression of thick clouds caused by unreasonable thresholds.

Next, we choose the optimal MFC and the suboptimal HOT to assess the efficiency of different strategies for pseudo labeling. Table VIII reports their accuracy and efficiency on the GF-1 training set, where "Time" means the time cost of pseudo labeling and "Mem" indicates the memory usage for the algorithm running. Producing even better annotations, MFC multiplies the time costs and computing resources of HOT. However, when using MFC *w/* RAPL as weak supervision, there is no evident improvement for HCDNet, and it remained comparable to HOT *w/* RAPL + HCDNet. These results indicate that RAPL can generate reliable pseudo cloud mask and BM with more specific semantic categories and spatial contours separately for HCDNet training. Moreover, HCDNet is capable of extracting representative features from these pseudo masks rather than relying on more accurate annotations.

VI. Conclusion

In this work, we propose a WDCD framework that leverages the physical rule to generate weak supervision for cloud detection. Specifically, an RAPL algorithm is devised to annotate pseudo cloud mask and BM with more explicit semantic categories and spatial structures separately based on cloud spectral properties without manual intervention. Then, these pseudo masks are taken as weak supervision to optimize the HCDNet training. Since clouds have complex geometric structures and nonuniform spectral reflectance, a DBRM is proposed to model the variable spatial transformations of clouds and activate sharp boundaries. Moreover, a harmonic loss is developed to recognize clouds with nonuniform spectral reflectance and suppress bright backgrounds. Extensive experiments demonstrate that the proposed method obtains the state-of-the-art performance.

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Yang Liu received the M.S. degree from Beijing Jiaotong University, Beijing, China, in 2019, where he is currently pursuing the Ph.D. degree with the School of Computer and Information Technology.

His research interests include computer vision, geospatial data mining, and remote sensing image understanding.

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Qingyong Li (Member, IEEE) received the B.Sc. degree in computer science and technology from Wuhan University, Wuhan, China, in 2001, and the Ph.D. degree in computer science and technology from the Institute of Computing Technology (ICT), Chinese Academy of Sciences (CAS), Beijing, China, in 2006.

He is currently a Professor with Beijing Jiaotong University, Beijing. He has more than 80 papers published and presented in prestigious conferences and journals, such as IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, SIGKDD, ICCV, CVPR, AAAI, ACM MM, and ICDM. His research interests include computer vision and artificial intelligence.



Xiaobao Li received the M.Sc. degree from the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo, China, in 2019. He is currently pursuing the Ph.D. degree with the School of Computer and Information Technology, Beijing Jiaotong University, Beijing, China.

His research interests include person reidentification, sparse representation, and deep learning.



Shuyi He received the B.Sc. degree from Beijing Jiaotong University, Beijing, China, in 2020, where she is currently pursuing the M.Sc. degree with the School of Computer and Information Technology.

Her research interests include computer vision, semantic segmentation, and remote sensing.



Fengjiao Liang received the B.Sc. degree from Shanxi University, Taiyuan, China, in 2016. She is currently pursuing the Ph.D. degree with the School of Computer and Information Technology, Beijing Jiaotong University, Beijing, China.

Her research interests include computer vision and image defect detection.



Zhigang Yao received the B.Sc. degree in atmospheric remote sensing and the Ph.D. degree in meteorology from the PLA University of Science and Technology, Nanjing, China, in 2001 and 2006, respectively.

He is currently a Senior Engineer with the Beijing Institute of Applied Meteorology, Beijing, China. He has published more than 50 articles. His research focuses on retrieving atmospheric and cloud parameters from active and passive instruments onboard satellites.



Jun Jiang received the M.Sc. degree in measuring and testing technologies and instruments from the PLA University of Science and Technology, Nanjing, China, in 2016.

He is currently an Engineer with the Beijing Institute of Applied Meteorology, Beijing, China. He has published more than ten articles. His research focuses on meteorological satellite radiometric calibration, cloud detection, and haze detection.



Wen Wang received the B.Sc. degree in information and computing science from Jilin University, Changchun, China, in June 2011, and the Ph.D. degree in computer science and technology from the Institute of Computing Technology (ICT), Chinese Academy of Sciences (CAS), Beijing, China, in June 2017.

From 2018 to 2020, she worked as a Post-Doctoral Researcher with ICT, CAS. She is currently a Lecturer with Beijing Jiaotong University, Beijing. Her research interests mainly include computer vision, pattern recognition, and machine learning, in particular, manifold learning, metric learning, deep learning, and their application in video-based face recognition.

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